

Early Years Schools Pollution Monitor

A Django-based real-time air quality monitoring platform
for primary schools and nurseries in Lambeth and Southwark

Python 3.13

Django 4.2

PostgreSQL 17

MIT License

Live Demo: <https://schools-air-quality-msp4-39fe66170249.herokuapp.com/>

Overview

Early Years Schools Pollution Monitor helps parents, school administrators, and policymakers make informed decisions about children's exposure to air pollution. The platform combines real-time sensor data from the London Air Quality Network (LAQN) and Breathe London with modelled pollution data (LAEI 2022) to provide comprehensive air quality information for **133 schools** across Lambeth and Southwark.

Key Objectives

- **Real-time monitoring:** Hourly updates from active sensors across two networks (LAQN and Breathe London)
- **Hybrid data approach:** Direct sensor readings where available, modelled data with real-time adjustments elsewhere
- **Tiered access:** Free map for all visitors; subscription-based premium features (£2.50/month via Stripe)
- **Transparency:** Clear data source attribution for every school

User Stories

User stories were developed using Agile methodology and are grouped by user type.

Parent / Family (Free Tier)

ID	As a...	I want to...	So that...	Status
US-01	Parent	View an interactive map showing all schools	See air quality at a glance	✓ Done
US-02	Parent	See colour-coded pollution levels on the map	Quickly identify good/poor air quality	✓ Done
US-03	Parent	Search for a school by name or postcode	Find my child's school quickly	✓ Done
US-04	Parent	Filter schools by borough	Focus on schools relevant to me	✓ Done
US-05	Parent	View detailed air quality data for a specific school	Understand pollution levels in detail	✓ Done
US-06	Parent	See where the data comes from (sensor vs modelled)	Judge how reliable the reading is	✓ Done
US-07	Parent	Register for an account	Access additional features	✓ Done
US-08	Parent	Add community notes about a school's environment	Share observations with other parents	✓ Done

Subscriber (Paid Tier — £2.50/month)

ID	As a...	I want to...	So that...	Status
US-09	Subscriber	Pay securely via Stripe	Access premium features safely	✓ Done
US-10	Subscriber	Manage my subscription status	See when it renews or cancel it	✓ Done
US-11	Subscriber	Edit school contact information	Keep details up to date	✓ Done
US-12	Subscriber	Edit and delete my own notes	Correct or remove my contributions	✓ Done

Site Administrator

ID	As a...	I want to...	So that...	Status
US-13	Admin	Access the Django admin panel	Manage all data and users	✓ Done
US-14	Admin	View and manage all school notes	Moderate community content	✓ Done
US-15	Admin	See which data source each school uses	Monitor data quality	✓ Done
US-16	Admin	View subscription and payment records	Handle customer support queries	✓ Done
US-17	User	Log in and log out securely	Account is protected	✓ Done
US-18	User	Reset my password if I forget it	Regain access to my account	✓ Done

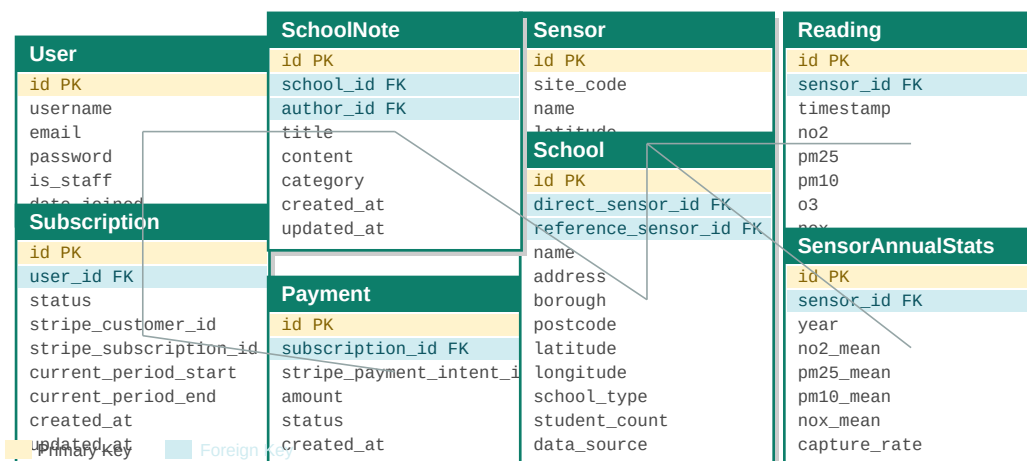
Implemented Features (MVP)

Interactive Map	Leaflet-based map displaying 133 schools and 42 sensors with colour-coded markers indicating pollution levels. Schools show popup cards with current readings, data source badges, and WHO guideline comparisons.
Tiered Data Sources	Three-tier approach: direct sensor readings for schools within 150m of a Breathe London sensor; LAEI baseline adjusted by real-time LAQN reference data for schools with a nearby reference sensor; static LAEI 2022 modelled data as fallback.
School Directory	Searchable, filterable list of all 133 schools with detail pages showing current air quality, WHO threshold comparison tables, and community notes.

Community Notes (CRUD)	Authenticated users can create, read, update, and delete notes about schools. Authors can only edit/delete their own notes. Categories: observation, concern, improvement, update.
Stripe Subscription	Full e-commerce flow using Stripe Checkout for £2.50/month subscription. Includes webhook handling for subscription lifecycle events and payment audit trail.
Authentication System	Django-based registration, login, logout, and password reset. Role-based permissions — staff users can edit school contact details, subscribers access premium features.
Responsive Design	Bootstrap 5 with custom CSS. Accessible navigation with skip links, ARIA labels, and semantic HTML. Footer with data source attribution links.

Database Schema

The application uses a relational database with **8 models** across **4 Django apps**. The schema centres on three domains: Schools, Sensors/Readings, and Users/Subscriptions.



Schools stores 133 primary schools and nurseries, each linked to up to two **Sensors** via *direct_sensor* (urban background sensor within 150m) and *reference_sensor* (nearest LAQN sensor for adjustment factors). Sensors produce **Readings** (hourly pollutant concentrations) and **SensorAnnualStats** (annual means used to calculate adjustment factors). **SchoolNotes** allow authenticated users to share community observations. The **Subscription** and **Payment** models handle Stripe-based access control, with a one-to-one link between User and Subscription.

Technology Stack

Backend	Python 3.13 · Django 4.2 · PostgreSQL 17 · psycopg2 · WhiteNoise
Frontend	Bootstrap 5 · Leaflet.js · Vanilla JavaScript
APIs / Data	LAQN (London Air Quality Network) · Breathe London via OpenAQ · LAEI 2022
Payments	Stripe — Subscription billing (£2.50/month)
Testing	Django TestCase · Coverage.py — 39/39 tests passing

Deployment

Heroku (PostgreSQL addon, Scheduler addon) · Git / GitHub

Data Sources & Tiered Approach

Tier	Source	Criteria	Confidence
Tier 1	Breathe London (direct)	School within 150m of sensor	High
Tier 2	LAEI 2022 + LAQN adjustment	Nearby LAQN reference sensor	Medium
Tier 3	LAEI 2022 static	No nearby sensor	Lower

Testing

39/39 tests passing — covering model creation, string representations, unique constraints, view responses, template rendering, JSON data structure, CRUD permissions, data source selection logic, and adjustment factor calculations.

```
python manage.py test          # All tests
python manage.py test schools  # Schools app
python manage.py test air_quality # Air quality app
python manage.py test maps    # Maps app
python manage.py test subscriptions # Subscriptions app
```

Deployment

The application is deployed to **Heroku** with the following configuration:

- Heroku app created with PostgreSQL addon
- Environment variables set via *heroku config:set*
- Procfile configured with *web: gunicorn schools_air_quality_msp4.wsgi*
- Static files served via WhiteNoise with CompressedManifestStaticFilesStorage
- Heroku Scheduler configured for hourly data fetches (:10 LAQN, :20 Breathe London)

```
git push heroku main
heroku run python manage.py migrate -a schools-air-quality-msp4
```

Security Notes

- **Stripe Integration:** The `@subscription_required` decorator is active on the `school_edit` view. Use test card **4242 4242 4242 4242** (any future expiry, any CVC).
- **API Key Management:** All sensitive keys are stored in environment variables via `.env` (local) and Heroku config vars (production). `.gitignore` prevents `.env` from being committed.
- **Lessons Learned:** The `.env` file was accidentally committed to git history early in development. Resolved by immediately rotating the Django `SECRET_KEY` and ensuring `.gitignore` was properly configured.

Production Checklist

- Set DEBUG=False in production
- Use production Stripe keys
- Configure ALLOWED_HOSTS with actual domain
- Enable HTTPS/SSL
- Set up monitoring and error tracking
- Configure regular database backups

Future Enhancements

- **Tiered Pricing:** Free tier for parents; paid tier for schools/administrators including historical trends, comparisons, and data export.
- **Air Quality Alerts:** Email and SMS notifications when pollution exceeds WHO thresholds, with recommended actions for schools.
- **Historical Trends:** Interactive graphs showing pollution over time, seasonal patterns, and year-on-year comparisons.
- **Data Export:** PDF reports for school governors, CSV downloads for researchers, and API access for third-party integrations.
- **Borough Expansion:** Scale from 2 boroughs to all 33 London boroughs. Migration to TimescaleDB recommended when exceeding 100,000 sensor readings.
- **Spatial Modelling:** Incorporate regression kriging and spatial regression techniques to control for environmental covariates — including proximity to busy roads, traffic volume, green space coverage, and building density — enabling more accurate interpolation of pollution surfaces between sensor locations and improved identification of the drivers of elevated exposure at the school level.

Author & Acknowledgments

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AI Assistance	GitHub Copilot · Claude (Anthropic) — brainstorming, debugging, and development support

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